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THE CHEMICAL COMPOSITION OF WILD BOAR (SUS SCROFA) MEAT HUNTED IN POLISH CARPATHIAN

Introduction:

The wild boar (Sus scrofa) is a highly adaptable species found in a variety of habitats (from semi-deserts to swamps and forests) and characterized by enormous reproductive potential (Colomer et al., 2024). The availability of food, the absence of natural enemies, climate warming (especially the lack of cold and snowy winters), and high reproductive potential have contributed to the rapid growth of this species' population. In Poland, wild boars are common throughout the country, from the Baltic coast to the mountainous regions. They are game animals and, although they play an important role in the forest ecosystem, their growing population and behavior can cause conflicts, especially in agriculture.

Results and Discussion:

Water content, Fat content, Protein content, Total ash content, Total carbohydrates content were presented in Table 1. Table 2 present CLA and UFA content of meat obrained from wild boars.

Table 1. Chemical composition, color parameters (L* a* b*) and cooking loss of *m. longissimus lumborum* of wild boars

Parameter	Subadults (1–2 years)	Adults (>24 months)
Chemical composition %		
Total solids	25.57±2.68	26.14±1.78
Protein	22.48±2.96	22.66±1.81
Fat	1.65±0.51	1.95±0.37
Ash	1.04±0.03	1.06±0.04
Carbohydrates	0.40±0.10	0.47±0.19
Caloric value kcal/100g	106.37±11.76	110.07±12.09
Color parameters:		
Lightness, L*	47.12±1.21	45.85±3.79
Redness, a*	18.23±1.35	19.47±1.97
Yellowness,b*	11.06±0.95	11.73±2.19
Cooking loss (%)	35.21±1.23[a]	36.72±1.89[b]
Cholesterol, mg 100g-1	50.35±4.12	52.08±3.75

The presented results of our own research and those of other authors show that meat composition, including the high fat and protein content, depends primarily on: food availability in hunting grounds, the age of the wild boar, carcass weight, and the hunting season. Compared to young animals, the meat of adult animals is characterized by higher fat and protein content and lower water content. The meat of wild boars hunted in late autumn and winter is characterized by a higher intramuscular fat content compared to the meat of wild boars hunted in spring and summer, which is due to the greater amount of food available from autumn crops. Among the analyzed meat components, the ash content is the least susceptible to changes and ranges from 0.9 to 1.3%

Conclusion:

Due to the growing wild boar population in Europe and the increased availability of wild boar meat resulting from culling aimed at depopulating this species, wild boar meat is becoming increasingly popular and more readily available on the market

Materials and Methods:

The aim of the study was to analyze the quality of meat from wild boars hunted in the Polish Carpathians - from the Bieszczady Mountains to the Silesian Foothills. The wild boars were obtained during depopulation hunts in the autumn-winter season of 2021-2022. Meat samples for analysis were collected at a game processing plant where the carcasses were sent after being gutted, skinned, and examined by a veterinarian. Raw meat (m. longissimus dorsi lumborum) from 20 carcasses of subadults (yearlings) wild boars (1–2 years) and 20 carcasses of adult wild boars (older than 2 years) was analyzed. The raw meat (*m. longissimus dorsi lumborum* between 13th and 14th rib) were minced and average samples obtained were subjected to following chemical analyses:

- Water content• Fat content
- Protein content• Total ash content
- Total carbohydrates content
- Fatty acid profile • aminoacid content

Table 2. Fatty acid profile of *m. longissimus lumborum* fat of wild boars

Fatty acids	Subadults (yearlings) 1–2 years	Adults (>24 months)
CLA	0.08±0.04	0.08±0.02
C-20:0	0.22±0.10	0.21±0.04
C-20:1*n-9*	0.79±0.31	1.08±0.44
C-20:2	0.19±0.06[a]	0.14±0.05[b]
C-20:3 *n-6*	0.03±0.01	0.02±0.004
C-20:4*n-6*	0.10±0,05	0.07±0.03
C-20:4*n-3*	0.05±0.02	0.03±0.02
C-20:5 *n-3*	0.01±0.004	0.01±0.003
C-22:4 *n-6*	0.03±0.01	0.03±0.01
C-22:5 *n-3*	0.04±0.02	0.03±0.01
C-22:6 *n-3*	0.02±0.008	0.02±0.003
PUFA *n-3*	0.38±0.05	0.34±0.03
PUFA *n-6*	4.30±1.90	4.07±1.27
n6/n3	11.19±3.05	11.97±2.04
PUFA	4.94±1.79	4.49±1.13
SFA	51.84±5.85	52.05±4.12
PUFA/SFA	0.095±0.01[a]	0.086±0.01[b]

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